University of California
CDL
California Digital Library

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Content Authenticity & Provenance

Engaging in the Library of Congress Government + LAM Community of Practice

CDL all-staff, April 2026

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UC3

C2PA + Library of Congress

Coalition for Content Provenance and Authenticity (C2PA)

- Build awareness of CAP for the government + LAM community and how this needs to evolve in light of AI
- Create informal community of practice/discussion about CAP concepts and potential implementations
- Collaborate on use cases and metadata templates
- Return feedback to C2PA to help the spec better reflect the needs of the G+LAM digital preservation community

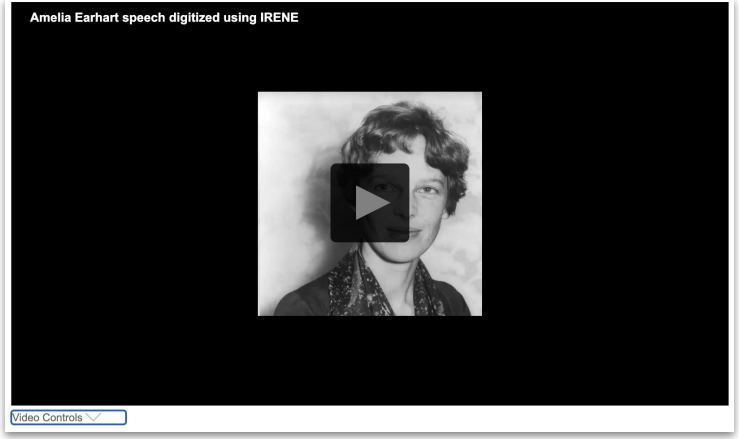
Content Authenticity & Provenance

TRUST AS CURRENCY

Content authenticity means verifying that digital content is genuine and unaltered. It does **NOT** mean that the content is factually correct.

Provenance is a verifiable (and trustworthy) history of digital content that shows its origin and any modifications it has undergone, including signal chain documentation.

FROM
Amelia Earhart's
International Broadcast
In
London
May 22
1932
As picked up in New York
by
SILVERTONE


Amelia Earhart speech digitized using IRENE
Video Controls

But now more than ever before…

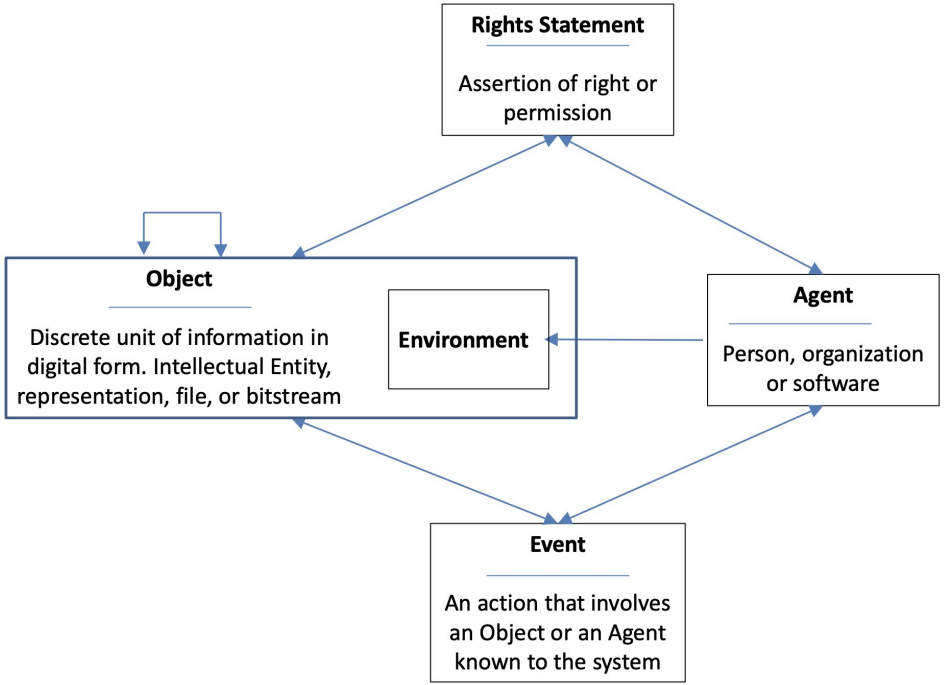
Authentic ≠ Factually Correct

METS RDA MARC21

Dublin Core PREMIS

Digital Preservation Actions

PREMIS



Levels of Digital Preservation

NDSA

Levels of Digital Preservation

Functional Area	Level			
	Level 1 (Know your content)	**Level 2 (Protect your content)**	**Level 3 (Monitor your content)**	**Level 4 (Sustain your content)**
Storage	Have two complete copies in separate locations Document all storage media where content is stored Put content into dedicated preservation storage	Have three complete copies with at least one copy in a separate geographic location Document storage and storage media indicating the resources and dependencies they require to function	Have at least one copy in a geographic location with a different disaster threat than the other copies Have at least one copy on a different storage media type Track the obsolescence of storage and media	Have at least three copies in separate geographic locations, each with a different disaster threat Diversify storage solutions to avoid single points of failure Plan for and execute actions to address storage hardware, software, and media obsolescence
Integrity	Verify integrity information if it has been provided with the content Generate integrity information if not provided with the content Virus check all content; isolate content for quarantine as needed	Verify integrity information when moving or copying content Use write-blockers when working with original media Back up integrity information and store copy in a separate location from the content	Verify integrity information of content at fixed intervals Document integrity information verification processes and outcomes Perform audit of integrity information on demand	Verify integrity information in response to specific events or activities Replace or repair corrupted content as necessary
Control	Determine the human and software agents that should be authorized to read, write, move, and delete content	Document the human and software agents authorized to read, write, move, and delete content and apply these	Maintain logs and identify the human and software agents that performed actions on content	Perform periodic review of actions/access logs
Metadata	Create inventory of content, also documenting current storage locations Back up inventory and store at least one copy separately from content	Store enough metadata to know what the content is (this might include some combination of administrative, technical, descriptive, preservation, and structural metadata)	Determine what metadata standards to apply Find and fill gaps in your metadata to meet those standards	Record preservation actions associated with content and when those actions occur Implement metadata standards chosen
Content	Document file formats and other essential content characteristics including how and when these were identified	Verify file formats and other essential content characteristics Build relationships with content creators to encourage sustainable file choices	Monitor for obsolescence, and changes in technologies on which content is dependent	Perform migrations, normalizations, emulation, and similar activities that ensure content can be accessed

This icon indicates recommendations with environmental sustainability considerations that may influence digital preservation decision-making. See the Environmental Sustainability Guide for further information.

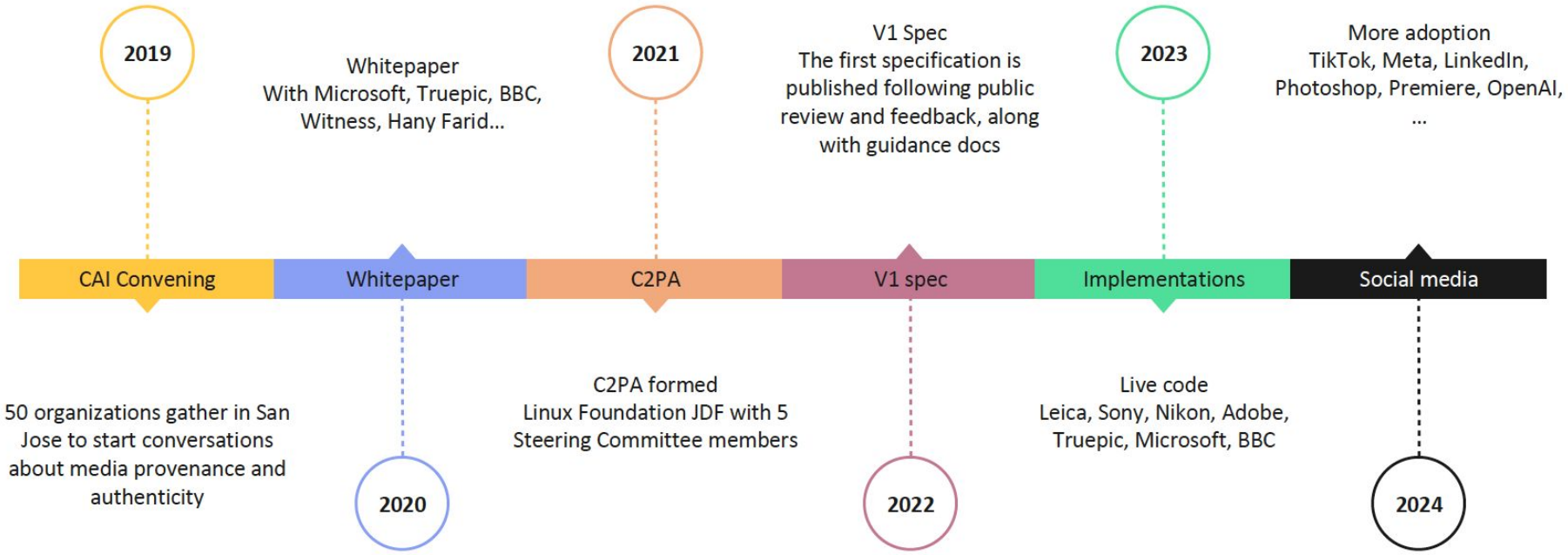
Levels of Digital Preservation v2.1

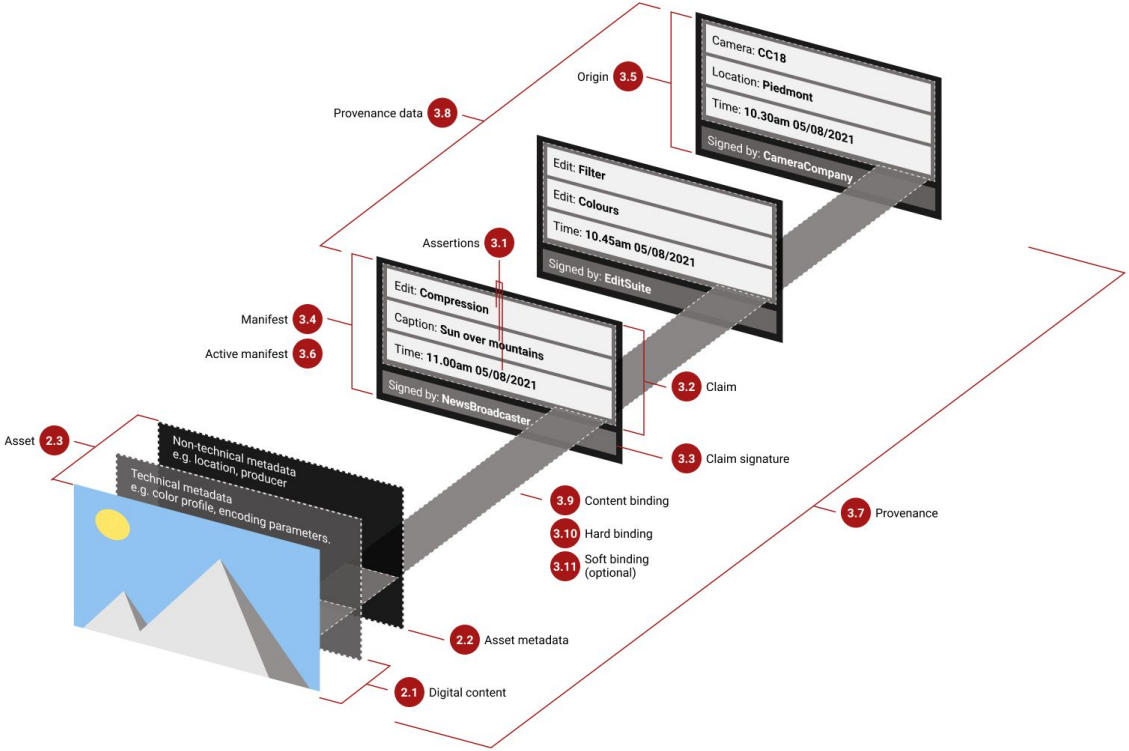
C2PA

Adobe BBC Google intel Meta Microsoft PUBLICIS GROUPE SONY OpenAI truepic amazon

Content Credentials & the Coalition for Content Provenance and Authenticity (C2PA)

- Steering committee members
- The C2PA spec is informed by scenarios, workflows and requirements gathered from industry experts and partner organizations
- Content Authenticity Initiative SDK


2019
Whitepaper
With Microsoft, Truepic, BBC, Witness, Hany Farid…
2021
V1 Spec
The first specification is published following public review and feedback, along with guidance docs
2023
More adoption
TikTok, Meta, LinkedIn, Photoshop, Premiere, OpenAI, …
CAI Convening
Whitepaper
C2PA
V1 spec
Implementations
Social media
50 organizations gather in San Jose to start conversations about media provenance and authenticity
2020
C2PA formed
Linux Foundation JDF with 5 Steering Committee members
2022
Live code
Leica, Sony, Nikon, Adobe, Truepic, Microsoft, BBC
2024



C2PA implementation diagram

Three C2PA Manifests make up the Provenance for this asset.

- Each manifest records an action, a.k.a. “assertion” in C2PA parlance
- Manifest data is cryptographically stored and digitally signed

This is all well and good, (mais)

"The report concludes that C2PA is a representation of authenticity and provenance, but (case in point) offers no guarantee of the truth or objectivity of the content itself, nor of the factual accuracy of the provenance claims within the manifest."

https://en.wikipedia.org/wiki/The_Treachery_of_Images#/media/File:MagrittePipe.jpg

The Treachery of Images
– René Magritte

Furthermore

from the Hacker Factor Blog, by Dr. Neal Krawetz:

"C2PA adds a cryptographic signature around the content and metadata. The thing to remember: a strong cryptographic signature around untrusted data does not make the data more trustworthy."

"...the detail, accuracy, and trustworthiness is dependent on the person who adds in the metadata."

And it's definitely "not a 'zero trust'," model, as is increasingly used in network security practices.

Enter: Trust as Currency

Back to the LC Community of Practice

TRUST AS CURRENCY

- User Stories
  - “How can C2PA identify and potentially limit access to content for AI model training based on Copyright status?”
  - “How can C2PA be leveraged to establish the provenance of Open Access publications governed by CC BY 4.0 license variations?”
- Action Assertion templates
  - A means to specify descriptive fields and controlled vocabulary related to a C2PA Assertion. e.g. **c2pa.cropped** `"description","softwareAgent","digitalSourceType"`

A Call to Action white paper

Content Authenticity and Provenance in the Age of Artificial Intelligence: A Call-to-Action for the LAMs Community

“In the face of rapid changes in technology and user expectations, LAMs must address how to ensure that digital collections content remains authentic, transparent, and verifiable from creation through access in order to meet the central community’s mission of public trust.”

Excerpts

How do we document the extent to which generative AI has “remixed” original information, such as a work of art or piece of literature? What do we even mean by “original” and “authentic” content?

Such questions must be addressed regardless of whether a digital object has come into contact with an AI system, as users will continue to expect the accuracy and verifiability of CAP data.

What is clear is that the emergence of AI technologies, especially generative AI, has introduced an entirely new dimension of authentication of LAMs collection content for consideration.

Excerpts

In digital preservation, authenticity is not only about maintaining bit-level stasis but about maintaining trustworthy documentation of an object's history….

Understanding this historical pattern clarifies why current debates around the relationship between CAP and AI are not unprecedented, but an extension of long-standing preservation concerns within a new technical terrain.

Excerpts

Whereas Web3 technologies such as blockchain offer a decentralized approach to authenticating digital assets, Model Context Protocol (MCP) considers the issue from the opposite vantage point.

MCP "is an open-source standard for connecting AI applications to external systems." Because MCP servers control access to collection data, they also preserve content integrity.

…there is assurance that requests for copies of digital objects are authentic versions, since MCP servers are limited to extracting information maintained by the host institution.

Excerpts

AI technologies, particularly generative and autonomous systems, threaten LAMs' capacity to maintain content authenticity not in some distant future but right now….

Without human review, there is a risk that automated provenance signals will be treated as authoritative endorsements rather than as structured disclosures (technical facts), leading users to place too much trust in AI-processed or algorithmically generated (CAP) representations of cultural heritage materials.

Four Pillars

Research and Development – Extend established CAP principles to new AI realities.

Partnerships and Collaboration – Deepen cross-institutional and cross-sector collaborations to avoid duplication, accelerate learning, and co-develop shared frameworks.

Advocacy with Industry, Vendors and User Communities – Actively shape standards by leveraging (LAMs) unique authority as long-standing stewards of public trust.

Open Distribution of Results and Lessons Learned – The rapid pace of AI innovation demands more transparent and collaborative modes of sharing experimental approaches and outcomes.

Wrap-up

“A societal transition is underway that is destabilizing our relationship to digital content. This shift will have profound effects on the LAMs community’s reputation as trusted repositories.”

Thank you! Questions?

merritt.cdlib.org/presentations

Eric Lopatin

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References

LC Community Running Notes
https://docs.google.com/document/d/1sODai27GpOb0vilICJkxYbCLpWN_AiI78u-Lr0uMOA4/edit?

Community blog post from July 2025:
https://blogs.loc.gov/thesignal/2025/07/c2pa-glam/

“Content Authenticity and Provenance in the Age of Artificial Intelligence: A Call-to-Action for the LAMs Community” white paper:
https://tinyurl.com/bdej7rd9

C2PA
Latest specification:
https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html
Spec Guidance for AI and ML:
https://spec.c2pa.org/specifications/specifications/2.4/ai-ml/ai_ml.html
Verify:
https://verify.contentauthenticity.org/
Conformance Explorer and Trust List
https://spec.c2pa.org/conformance-explorer/

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User stories submitted to C2PA :
https://docs.google.com/document/d/1f6-cN7G4ZF66VXThy8Bdl4UZD_GXn5vldROAyRAscIk/edit?tab=t.0#heading=h.v24gtpgmqr62

Content Credentials
https://contentcredentials.org/

Content Authenticity Initiative
https://contentauthenticity.org/
https://github.com/contentauth

Trust In Archives
https://www.trustarchives.org/

Digital Object Authenticity Working Group (DOAWG)
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Leonard Rosenthol's Jan. 2025 preso:
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